Exponential Growth & Decay

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Exponential Growth & Decay

The ideas of compound interest and depreciation can be applied to other (non-money) situations, such as increasing or decreasing populations.

What is exponential growth?

- When a quantity **grows exponentially** it is **increasing** from an original amount by a scale factor (or percentage) each year for x years
  - Some questions use a different timescale, such as each day, or each minute
- Real-life examples of exponential growth include:
  - Population increases
  - Bacterial growth
  - The number of people infected by a virus

What is exponential decay?

- When a quantity **exponentially decays** it is **decreasing** from an original amount by a scale factor (or percentage) each year for x years
  - Some questions use a different timescale, such as each day, or each minute
- Real-life examples of exponential decay include:
  - The temperature of hot water cooling down
  - The value of a car decreasing over time
  - Radioactive decay (the mass of a radioactive substance over time)

How do I use models for exponential growth or decay?

- Scenarios which exponentially grow or decay can be **modelled with an equation**
- A useful format for this equation is
  - $y = a^x$
    - E.g. $y = 4^x$ or $y = 0.8^x$

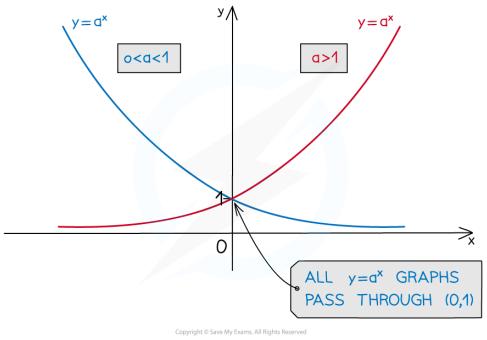
Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $y = 3^x$ could model the number of flies, y, in a population of flies, against the number of days, x
  - The initial number of flies, when $x = 0$, is $3^0 = 1$
  - After 2 days, the number would be $3^2 = 9$
  - After 4 days, the number would be $3^4 = 81$
  - From the values above, and the fact that $a > 1$, this is **exponential growth**

- $y = 0.5^x$ could model the radius of a water droplet, y cm, as it evaporates where x is measured in minutes
  - The initial radius of the droplet, when $x = 0$, is $0.5^0 = 1$ cm
  - After 2 minutes, the radius would be $0.5^2 = 0.25$ cm
  - After 4 minutes, the radius would be $0.5^4 = 0.0625$ cm
  - From the values above, and the fact that $0 < a < 1$, this is **exponential decay**

How are exponential models related to exponential graphs?

- Plotting an **exponential model**, $y = a^x$, on a set of axes gives an **exponential graph** with the following shape



- The **y-intercept** is always 1
  - This is because the y-intercept is where $x = 0$
  - $y = a^0 = 1$

- There is an **asymptote** at $y = 0$ (the curve gets closer and closer to the x-axis without crossing it)
  - This is because there are no solutions to $0 = a^x$

Are there any other exponential models I may encounter?

- You may see exponential models in the form $y = ba^x$
  - The coefficient, b, scales (**stretches**) the graph **vertically**
  - The y-intercept will be $(0, b)$
    - This is because when $x = 0$, $y = ba^0 = b(1) = b$
  - b is the **initial** (starting) **amount** of the quantity being modelled

answer

- It may want the final amount to the nearest thousand or to the nearest integer

Worked Example

A large population of birds increases exponentially according to the model

$$N = 1.04^t$$

where N is the number of birds, measured in thousands, and t is the number of years.

Find the population after 13 years, giving your answer to the nearest hundred.

Substitute in $t = 13$

$$N = 1.04^{13}$$
$$N = 1.665073507...$$

The question asks for the answer to the nearest hundred
The question also states that N is measured in thousands

1.665073507... × 1000 = 1665.073507... birds

Round to nearest hundred

1700

Worked Example

The percentage of tickets still available for a concert is modelled approximately by the equation

$$y = a^x$$

Where y is the percentage of tickets still available, as a decimal, and x is the number of days since the tickets have been released.

(a) After 3 days, 51.2% of the tickets are still available. Find the value of a.

x will be 3 (from 3 days)
y will be 0.512 (as the percentage is written as a decimal in this model)
Substitute these values into the model

$$0.512 = a^3$$

Find a by taking the cube root of both sides

$$\sqrt[3]{0.512} = a$$
$$0.8 = a$$

$$a = 0.8$$

Now that a is known, the full equation of the model is

$$y = 0.8^x$$

x is measured in days

2 weeks = 2 × 7 = 14 days

Substitute in $x = 14$

$$y = 0.8^{14} = 0.04398046511...$$

Turn this into a percentage by multiplying by 100

4.398046511... %

Round to three significant figures

4.40 %

(c) State one reason why this model may not be appropriate for the scenario.

The value of y can never reach zero (there is an asymptote at $y = 0$).
The model suggests that the tickets never truly run out (which is unrealistic).

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